

Stakeholder Engagement Plan

Sesotho-English Agricultural Advisory Small Language Model for Smallholder Farmers in Lesotho

1. Stakeholder Groups Selected

Drawing on our stakeholder map (built using Wakunuma and Jiya's method), we selected two groups whose involvement is critical to the responsible development of our agricultural advisory model:

- Women farmers' community groups — representing women smallholders across all four agro-ecological zones, who make many planting and post-harvest decisions but are historically underserved by extension systems.
- Agricultural extension officers (Ministry of Agriculture) — the domain experts who currently deliver advice, who will be most affected by the model's introduction into their workflow.

2. What Matters Most to Each Group

Women farmers' groups care most about safe, accessible information in Sesotho that fits their daily realities — household gardens, poultry, and post-harvest handling — without requiring smartphone ownership, data bundles, or travel. They value cultural preservation, control over their own knowledge and data, and tools that strengthen rather than bypass their existing community networks.

Extension officers care most about professional credibility and workload: the model must be agronomically accurate, cite verifiable sources, and route difficult cases to them rather than erode their role. They also value tools that extend their reach in under-staffed districts and respect the trust relationships they have built with farmers over years of service. Both groups want a system that is inclusive, trustworthy, and grounded in local realities.

3. Potential Harms and Benefits

- **Potential harms:**
- Exclusion or misrepresentation of women farmers if their questions, dialects, and farming contexts are missing from the dataset.
- Reinforcement of gender stereotypes if the model defaults to 'male farmer' assumptions or dismisses women's agricultural domains as marginal.
- Erosion of extension officers' expertise and authority if the model contradicts or bypasses them, or if farmers stop seeking their judgment.
- Loss of trust if the model gives wrong advice that officers are then blamed for.

- **Potential benefits:**
- Women's empowerment through timely, independent access to verified advice in their own language.
- Extension officers gain a triage tool that multiplies their reach and frees time for complex cases.
- Co-created datasets that preserve Sesotho agricultural vocabulary and indigenous knowledge.
- Stronger, more accountable links between communities, officers, and the Ministry.

4. Engagement Methods Selected (from the Action Catalogue)

- Women farmers' groups: Distributed Dialogue — facilitated conversations held within existing community structures across multiple districts over time.
- Extension officers: Participatory Design — co-design workshops where officers contribute examples, define the query taxonomy, and review model outputs.

5. Why Each Method Fits

Method 1: Distributed Dialogue with Women Farmers' Groups

Distributed Dialogue is an engagement method in which facilitated conversations are spread across multiple settings and times, carried by trusted community structures rather than a single centralized event. For women farmers in Lesotho, this method fits our context better than any workshop-based alternative, for several reasons.

First, geography and access. Our stakeholders live scattered across four agro-ecological zones, from lowland villages to remote mountain districts such as Mokhotlong and Thaba-Tseka. Expecting women smallholders — many of whom lack smartphones, transport, or free time — to attend a central workshop would reproduce exactly the exclusion our project exists to address. Distributed dialogue instead brings engagement to them, through existing gatherings such as village savings groups, church groups, and cooperative meetings, where women already meet and feel safe speaking.

Second, trust and power. Research on engagement in African contexts shows that women often defer to male voices in mixed formal settings, particularly where agricultural decision-making is concerned. Separate women's dialogues, facilitated in Sesotho by female facilitators from their own districts, create the conditions for honest input about what advice they actually need, what language they actually use, and what they fear about an AI tool. This directly responds to the risk — demonstrated by the Nigerian chatbot example in our course materials — that excluding women results in systems that misrepresent or stereotype them.

Third, depth over time. Distributed dialogue is iterative: conversations in one district inform the questions asked in the next, and early findings can be taken back to earlier

groups for validation. This suits our dataset lifecycle, where consent, vocabulary, and use-cases evolve across collection rounds. It also builds durable relationships: the women who host and shape these dialogues become natural members of our community advisory panel and our grievance feedback network, keeping accountability alive after launch.

Finally, cultural fit. Sesotho community decision-making traditionally values dialogue carried in familiar settings and in the local language, not extractive one-off surveys. Choosing this method signals respect for those norms — aligning with our value of community ownership rather than merely consulting about it.

Method 2: Participatory Design with Extension Officers

Participatory Design is a co-creation method in which the people who will use or be affected by a system design it alongside the developers — contributing concrete examples, defining categories, and evaluating prototypes. For extension officers, whose professional role the model will reshape, this method is essential rather than optional.

First, it converts potential adversaries into co-owners. Officers' most legitimate fear is displacement: a tool that answers farmers directly could be perceived as replacing them, and an undermined extension service would destroy the trust channel our whole model depends on. Co-design workshops where officers define the question taxonomy (planting, pests, fertilizer, livestock, weather, out-of-scope) turn their expertise into the system's backbone. The model then amplifies their professional judgment rather than competing with it — routing difficult and regulated cases to them by design, because they designed that routing.

Second, it is the strongest available accuracy mechanism. Officers hold the tacit agronomic knowledge — local pest outbreaks, which lowland advice fails in the mountains, when a generic answer becomes dangerous — that no document captures. In workshops, officers contribute real anonymized query logs, label edge cases, and review draft model outputs against their field experience. This grounds our dataset in verified practice, not just published sources, and creates the gold-standard label set described in our Data Card.

Third, it builds institutional legitimacy and sustainability. A model developed with the Ministry of Agriculture's officers has an owner inside the system that will outlive any grant. Their involvement eases permissions for data (advisory leaflets, historical records), creates a natural channel for the monthly output audits we committed to in our Impact Statement, and aligns the tool with national agricultural policy rather than a one-off pilot.

Finally, participatory design fits our values explicitly: it operationalizes community ownership and trustworthiness. When officers can say 'we helped design this,' farmers' trust transfers to the tool — and our accountability loop (advisory panel, SMS grievance channel, annual Sesotho-language reviews) has expert partners inside it from day one.

6. Influence on Dataset and Model Design

The distributed dialogues will directly shape dataset composition: at least 50% women participants, women's focus groups to prevent male voices dominating, and Sesotho question variants capturing household-garden, poultry, and post-harvest domains that officers' logs miss. Consent will be oral and in Sesotho, with district-level anonymization. The co-design workshops will define the routing taxonomy, the gold-standard label set, and the mandatory source citation format in every answer. Together, the two methods ensure the model's outputs are linguistically authentic, agronomically verified, and routed to escalate — rather than guess — whenever human judgment is required.

References

- [1] Wakunuma, K. and Jiya, T. (2019). Stakeholder Engagement and Responsible Research & Innovation in promoting Sustainable Development and Empowerment through ICT. *European Journal of Sustainable Development*, 8(3), 275.
- [2] Gendered Chatbots in Nigeria: Critical Perspectives. In *Responsible AI in Africa* (2022), Springer, pp. 119-139.
- [3] The Engage2020 Action Catalogue (2015). <https://actioncatalogue.eu/>